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Coherence in the brain unfolds across separable temporal regimes

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Abstract

Coherence in language requires the brain to satisfy two competing temporal demands: gradual accumulation of meaning across extended context and rapid reconfiguration of representations at event boundaries. Despite their centrality to language and thought, how these processes are implemented in the human brain during naturalistic listening remains unclear. Here, we tested whether these two processes can be captured by annotation-free drift and shift signals and whether their neural expression dissociates across large-scale cortical systems. These signals were derived from a large language model (LLM) and formalized contextual drift and event shifts directly from the narrative input. To enable high-precision voxelwise encoding models with stable parameter estimates, we densely sampled one healthy adult across more than 7 hours of listening to thirteen crime stories while collecting ultra high-field (7T) BOLD data. We then modeled the feature-informed hemodynamic response using a regularized encoding framework validated on independent stories. Drift predictions were prevalent in default-mode network hubs, whereas shift predictions were evident bilaterally in the primary auditory cortex and language association cortex. Furthermore, activity in default-mode and parietal networks was best explained by a signal capturing how meaning accumulates and gradually fades over the course of the narrative. Together, these findings show that coherence during language comprehension is implemented through dissociable neural regimes of slow contextual integration and rapid event-driven reconfiguration, offering a mechanistic entry point for understanding disturbances of language coherence in psychiatric disorders.

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Introduction

Coherence is a defining property of language and thought: meanings accumulate across time, remain constrained by context, and yet must be flexibly updated when situations change. This capacity underpins narrative comprehension and reasoning and is profoundly disrupted in psychopathology, most notably in formal thought disorder [1–4]. Despite its centrality, coherence remains poorly specified at the neural level. Classical neurobiological models of language emphasize local linguistic operations within a largely left-lateralized frontotemporal system [5–7], while naturalistic fMRI studies highlight distributed semantic representations across association cortex [8–11]. More recently, deep neural networks have been shown to robustly predict brain responses during continuous language comprehension [12–15]. Yet even within this emerging framework, coherence itself remains an underspecified construct.

At a computational level, maintaining coherence entails at least two distinct temporal operations. Meaning must be integrated gradually across extended context, producing smooth representational drift over seconds to minutes [9, 11, 16, 17]. At the same time, context must be rapidly reconfigured at event boundaries, when shifts in topic or scene render prior interpretations obsolete [18–21]. Neural correlates of these operations appear partially dissociable, with default-mode and temporoparietal networks exhibiting long integration windows and sensitivity to higher-order narrative structure [9, 10, 16, 22, 23] and auditory and language-selective cortex showing rapid, boundary-sensitive responses [24–26]. However, these temporal regimes are typically examined in isolation, using incompatible paradigms and analytic frameworks, leaving unresolved how gradual integration and discrete reconfiguration jointly implement coherence during naturalistic comprehension [27–30].

Progress on this problem has been constrained by the absence of stimulus-derived signals that jointly capture continuous context accumulation and discrete reconfiguration. Data-driven segmentation methods applied to neural activity can identify when large-scale reorganization occurs, but remain agnostic to the computational signals driving these transitions [20, 21, 28, 30]. Conversely, encoding models have largely emphasized local linguistic structure, offering limited access to discourse-level context and event structure [8, 31]. Recent advances in large language models provide a way to bridge this divide. By maintaining internal representations that evolve with accumulated context and change sharply at narrative boundaries, these models enable annotation-free formalization of contextual drift and event shifts directly from naturalistic input [12, 13, 15, 32]. Combined with voxel-wise encoding and dense-sampling fMRI, this approach allows a unified test of how distinct temporal regimes jointly support coherence across cortical systems.

Here, we combined model-derived drift and shift signals with high-precision fMRI during extended narrative listening. We densely sampled a single healthy participant across multiple hours of stories, enabling stable voxel-wise encoding models and out-of-sample validation at the level of entire narratives (Figure 1). We hypothesized

that drift signals would be expressed more strongly in association networks with long integration windows, including default-mode and parietal regions, whereas rapid shift signals would preferentially engage auditory and language-selective regions sensitive to boundary-related reorganization. By partitioning explanatory variance between these processes within a single model-based framework, this work advances a mechanistic account of coherence in language comprehension and provides a principled foundation for probing coherence disturbances in psychiatric conditions.

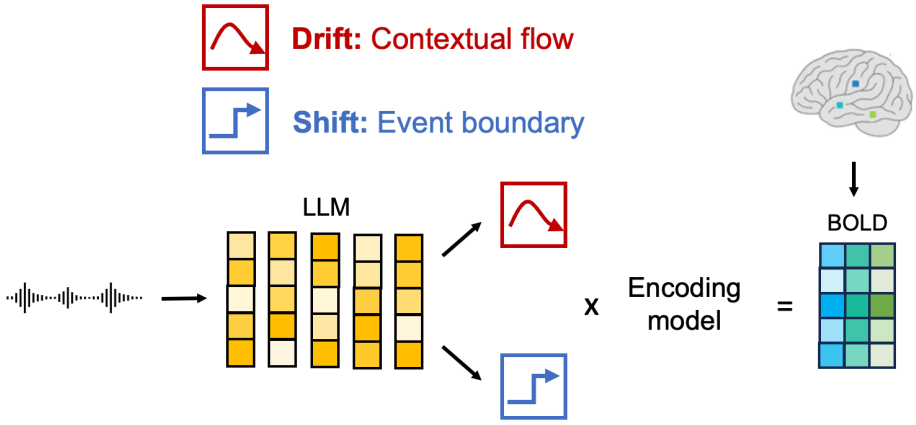

Figure 1: Annotation-free mapping of narrative coherence to brain dynamics. Crime stories were processed by a decoder-only large language model (LLM) to derive two complementary signals of narrative structure: *drift*, capturing the gradual accumulation of contextual meaning, and *shift*, capturing discrete event boundaries. These LLM-derived signals were aligned to the spoken narratives presented during fMRI and entered into an encoding model to predict voxelwise BOLD responses. Comparing predicted and observed activity allowed us to map distinct neural systems associated with gradual contextual integration and rapid event-driven reconfiguration across the cortex.

Results

Dissociable cortical systems encode drift and shift

Drift and shift signals exhibited dissociable whole-brain predictive topographies, with drift effects distributed across the heteromodal association cortex, and shift effects concentrated in peri-Sylvian language regions (Figure 2). To evaluate how well these effects generalized across narratives, we evaluated cross-story consistency by region of interest. For this, we used region of interest-level *Simes counts*, defined as the number of stories (out of 13) in which a region showed a significant effect (Materials and Methods). Figure 2 summarizes these counts across the Harvard–Oxford atlas [33]. More specifically, for *drift*, effects showed that replicability concentrated in heteromodal association hubs. The most consistent effects were observed in angular gyrus, precuneus, posterior cingulate, and superior parietal regions, whereas early auditory cortex rarely showed replicable drift effects under identical criteria. Thus, drift generalized preferentially in default-mode and parietal hubs rather than in auditory

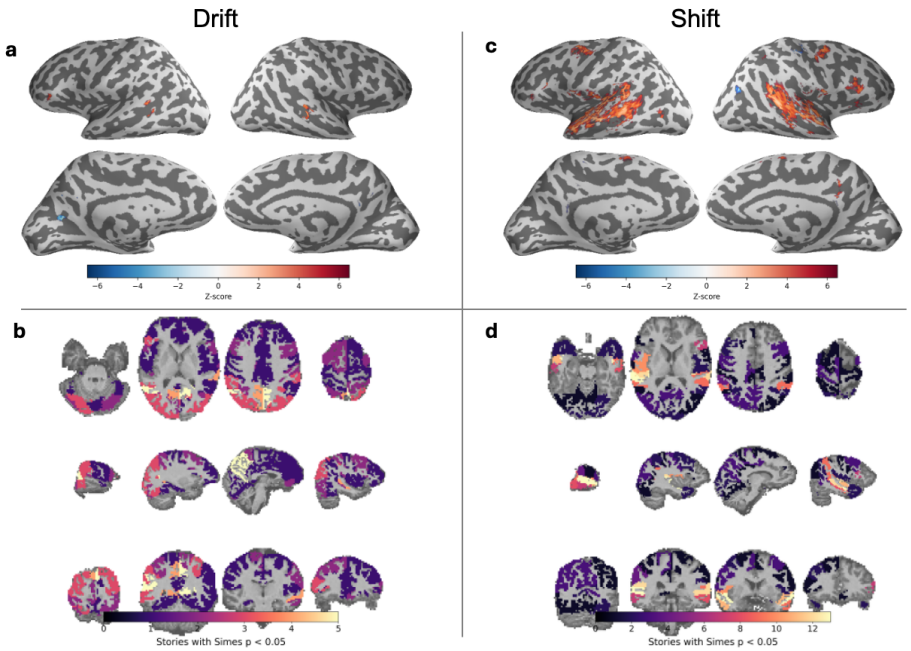

Figure 2: Across stories activation and consistency maps for drift and shift. **a)** Cortical surface projections showing Z-scores for drift. **b)** Volumetric region of interest maps showing the cross-story consistency of effects, where color intensity represents the number of stories (out of 13) in which each region was significant (Simes $P < 0.05$). The drift maps showed comparatively weaker and more distributed patterns with lower consistency, involving higher-level hubs like the angular gyrus and precuneus (default mode network), consistent with integrative or attentional components rather than primary auditory drive. **c)** Cortical surface projections displaying Z-scores for shift. **d)** The shift maps exhibited robust, bilateral modulation within classic speech areas around the Sylvian fissure (e.g., Heschl's gyrus and STG), showing high consistency across up to 13 stories.

core regions (Figure 2; Table S1). By contrast, *shift* effects showed a complementary profile. They generalized robustly across narratives within peri-Sylvian speech regions. Core auditory–language regions reached ceiling or near-ceiling consistency, including Heschl's gyrus, planum temporale/polare, and superior temporal cortex bilaterally. These patterns indicate highly reproducible, boundary-linked responses to shift across stories (Figure 2).

Unique contributions of drift and shift

Because drift and shift are correlated, we estimated their *unique* contributions by fitting a joint model and testing voxelwise coefficients across held-out stories (Materials and Methods). Results are summarized by voxelwise significance maps, region-level effect sizes, and the directional contrast $\Delta\beta = \beta_{\text{shift}} - \beta_{\text{drift}}$ (Figure 3).

Drift shows unique effects in heteromodal association hubs

Unique drift effects localized to higher-order association regions, most consistently in angular gyrus, precuneus, posterior cingulate, and medial prefrontal cortex (Figure 3).

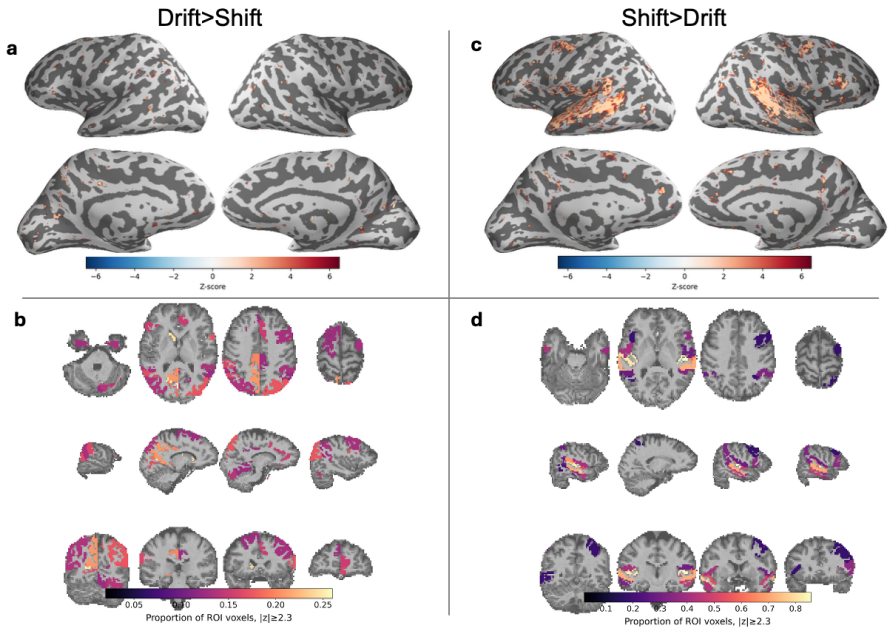

Figure 3: Unique predictive contributions of drift and shift. a) Voxelwise significance map for unique drift effects, localized primarily to heteromodal association hubs, including angular gyrus and precuneus. **b)** Corresponding ROI-level effect sizes for drift in posterior-inferior default-mode regions (DMN–PI). **c)** Voxelwise significance map for unique shift effects, estimated after regressing out shared variance with drift. Effects concentrate bilaterally in the peri-Sylvian language network, including Heschl's gyrus, superior temporal cortex, and planum temporale/polare. **d)** ROI-level effect sizes for shift in language-network ROIs (LANG), showing mean unique regression weights (β) across 13 stories with 95% bootstrap confidence intervals.

In these hubs, the directional contrast $\Delta\beta$ was small, consistent with drift contributing reliably but not overwhelmingly relative to shift.

Shift uniquely explains peri-Sylvian language responses

Unique shift effects concentrated bilaterally in the peri-Sylvian language belt, including Heschl's gyrus, planum temporale and polare, anterior and posterior superior temporal cortex, posterior supramarginal gyrus, and inferior frontal gyrus (pars triangularis) (Figure 3). Accordingly, $\Delta\beta$ was positive throughout these regions, indicating that shift explains explanatory variance beyond drift within canonical speech-processing cortex.

Secondary drift-dominant effects outside the language system

Outside the language network, several sensory–motor and visual regions exhibited negative $\Delta\beta$ values (drift > shift), including somatosensory cortex and occipital areas, indicating stronger unique contributions of slowly varying contextual signals in these processing streams (Table S2).

Effect sizes and coefficient sign

To quantify magnitude beyond significance, we report mean unique regression weights across stories with 95% bootstrap confidence intervals for representative ROI sets (DMN–PI for drift, LANG for shift; Figure 3). Because predictors and BOLD signals were z-scored within story and modeled with FIR lags, coefficient sign reflects relative phase alignment rather than excitation versus inhibition; interpretation therefore focuses on spatial distribution and relative magnitude of unique effects.

Drift depends on a finite integration timescale

To characterize the temporal window over which contextual information contributes to drift, we varied the memory decay parameter ρ and recomputed drift predictors across a wide range of integration timescales. For each value of ρ, we evaluated cross-story replicability using the same voxel-wise inference and ROI-level Simes aggregation as in the main analyses (Materials and Methods).

Across timescales, drift effects generalized more strongly in posterior-inferior default-mode and parietal regions (DMN–PI) than in the language network (LANG) (Figure 4). Replicability peaked at intermediate integration timescales ($\rho \approx 0.10$–0.20), whereas very short timescales yielded near-memoryless traces with weak generalization and very long timescales over-smoothed the signal. Thus, drift reflects integration over a finite temporal window, rather than unlimited accumulation, with a preferred timescale in the heteromodal association cortex.

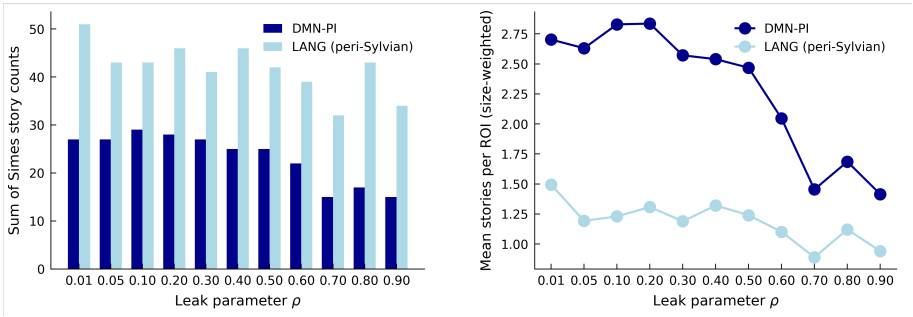

Figure 4: Replicable drift effects across integration timescales. Cross-story generalization of drift effects as a function of the memory decay parameter ρ **a)** Raw sums of region of interest-level Simes significant-story counts (not normalized for ROI number or size). **b)** Size-weighted mean significant stories per region, normalizing for both region number and size. *After normalization*, drift generalized more strongly in posterior-inferior default-mode and parietal regions (DMN–PI) than in the language network (LANG), with a broad maximum at intermediate integration timescales ($\rho \approx 0.10$–0.20).

Punctuation control: shift predicts boundary responses beyond pauses

To test whether the shift signal merely reflects low-level pauses or typography, we included punctuation (`.,:;`) as a competing predictor. When entered into the joint

model, shift retained large and focal unique effects across bilateral peri-Sylvian regions, including Heschl's gyrus, planum temporale/polare, and superior temporal cortex, whereas punctuation showed markedly smaller unique effects in the same areas (Figure S5). Moreover, punctuation exhibited limited cross-story generalization, reaching significance in only a small subset of ROIs and stories, in contrast to the robust and widespread replicability observed for shift. These results indicate that shift captures boundary-linked neural responses that cannot be explained by pauses or typographic markers alone.

Summary

Together, these analyses showed that coherence during narrative comprehension is implemented through two separable but interacting temporal regimes: a context-integrating drift signal expressed in the heteromodal association cortex with a finite preferred timescale, and a boundary-linked shift signal that generalizes robustly across peri-Sylvian language regions.

Discussion

In this study, we demonstrated that coherence during naturalistic language comprehension can be decomposed into two temporally distinct but co-expressed processes: a context-integrating signal tracking gradual semantic evolution (drift), and boundary-linked signal marking event transitions (shift). Both signals were derived annotation-free from a single large language model and predicted BOLD responses during extended story listening, yet they dissociated systematically across cortical systems. Drift generalized preferentially in heteromodal association hubs, including default-mode and parietal cortex, whereas boundary sensitivity was dominant in peri-Sylvian auditory–language regions. Crucially, unique-effects analyses revealed that these processes are not mutually exclusive: higher-order hubs such as angular gyrus and precuneus expressed sensitivity to both signals, suggesting that coherence emerges from their coordinated interaction rather than from a single mechanism.

These findings situate coherence at the intersection of two lines of work that have often been treated separately: gradual context integration and event segmentation [18–21, 27]. By formalizing both operations within a single, stimulus-derived framework, our results reconcile perspectives that emphasize discrete event boundaries with those that stress continuous representational drift. The observation that default-mode hubs track both drift and shift aligns with proposals that these regions maintain situation models across time while remaining sensitive to structural changes that delimit events. Our results also map naturally onto hierarchical temporal receptive window (TRW) accounts of naturalistic comprehension, which posit progressively longer integration windows from early auditory cortex to association cortex [9, 16, 17, 34]. Within this framework, rapid boundary-linked responses are expected to peak in peri-Sylvian speech regions, whereas slower accumulation of contextual meaning should preferentially engage posterior and medial association hubs. Consistent with this view, drift effects generalized most strongly in posterior-inferior default-mode and parietal regions and peaked at intermediate integration timescales, indicating that these regions integrate

context over a finite window rather than accumulating information indefinitely. This timescale tuning provides a concrete instantiation of TRW hierarchies using explicit, model-based predictors [11, 25].

Recent work has shown that transformer-based language models robustly predict neural responses during language comprehension and align hierarchically with cortical processing stages [12–14]. Importantly, such alignment does not imply that next-token prediction alone captures human comprehension [15]. Our results extend this literature by showing that LLM-derived internal dynamics can be used to formalize distinct temporal components of coherence itself. In particular, recent studies demonstrate that large language models can segment narrative events in a manner comparable to human judgments [32], providing an external validation for the use of LLM-based shift signals as proxies for narrative structure.

Data-driven event segmentation approaches applied directly to neural activity, such as hidden Markov models, have reported hippocampal responses time-locked to inferred event boundaries [20, 21, 35]. We did not observe reliable hippocampal effects in the present data. This absence may reflect the temporal sparsity of hippocampal boundary responses, their sensitivity to memory encoding and retrieval demands, or technical challenges associated with subcortical measurement at ultra-high field. Targeted acquisition strategies and analyses explicitly probing memory updating or reinstatement may be required to clarify hippocampal contributions to coherence during naturalistic comprehension [36].

Event boundaries in speech are often accompanied by prosodic cues such as pauses, pitch resets, and durational changes, which elicit robust responses in early auditory cortex [37–39]. Neurophysiological and neuroimaging studies show that such acoustic edges drive strong boundary-locked responses in Heschl's gyrus and superior temporal cortex, even in the absence of lexical content [26, 40–42]. Our control analyses showed that a punctuation-based regressor captured a limited portion of this variance, but that the LLM-derived shift signal explained substantial, focal variance beyond punctuation alone. This dissociation indicates that shift captures higher-order boundary structure rather than merely reflecting pauses or typographic markers.

Several limitations warrant consideration. First, our dense-sampling design focused on a single participant and a single language and genre, enabling high precision but limiting generalizability. Second, acoustic and prosodic features were not explicitly modeled and may account for residual variance in early auditory regions. Third, implementation choices, including language-model architecture and parameterization, were principled but not unique. Future work should examine how drift and shift representations depend on model family, scale, and layer, and how they generalize across participants, languages, and stimulus types.

Despite these limitations, the present framework provides a principled way to link neural activity to explicit, time-resolved computational constructs of language processing. Because both drift and shift signals are derived automatically from the stimulus, without manual annotation, they offer scalable assays of contextual integration and segmentation. Such measures may prove useful for probing disturbances of discourse organization, such as those described as formal thought disorder, a core feature of psychosis. More broadly, this work illustrates how model-based feature spaces can be used to partition explanatory variance across temporal regimes, advancing a mechanistic

account of how the brain maintains and reorganizes meaning over time.

Materials and Methods

Sample size and scanning schedule

One healthy, right-handed, male, native German speaker (age=28 years) was densely sampled across eight scanning sessions spanning 43 days (16 April–29 May 2024). Across all sessions we collected 16 narrative runs; 3 runs were excluded a priori (1 due to excessive motion; 2 due to signal loss), leaving 13 runs for analysis (one story per run). Exact dates, start times, run counts, and story assignments are listed in Table S3.

Stimuli

The participant listened passively to 13 narrated, single-voice German renditions of classic crime short stories (one per run) from the 19th/early 20th century (median 31.5 min; range 13.8–55.0 min). Full titles, authors, and per-scan assignments are listed in Table S3. Audio was delivered via MR-compatible insert earphones (Optoacoustics FOMR III+). Each run contained one continuous story (no task). We used leave-one-story-out cross-validation at the level of stories (each story serving once as the held-out test set; Supplementary Methods). To reduce onset/offset transients we trimmed training stories by 20 TRs at both onset and offset and test stories by 30 TRs at onset and 20 TRs at offset.

Text-audio alignment (overview)

We force-aligned the story text to audio at the word level using the `torchaudio` MMS pipeline [43, 44] on romanized/normalized text, producing word start/end times (Praat format). Text-derived features were converted to continuous time and resampled to the TR grid. Full alignment and resampling details (including the resampling kernel) are provided in g.

LLM-derived signals (overview)

We derived two *annotation-free* signals from the same decoder-only LLM (Llama-3.3-70B-Instruct[45]). *Shift* is the per-word log-probability that the next generated symbol is a special boundary marker; *drift* is the magnitude of change in a leaky-integrated hidden-state trace. Both signals were aligned to word times, resampled to TR, z-scored within story, and expanded into a 14-lag Finite Impulse Response (FIR) design. The exact prompt, layer/windowing choices, drift definition, and the boundary feature bank are given in Supplementary Methods.

Data acquisition and preprocessing (overview)

Data were acquired on a 7T Siemens MAGNETOM Terra with a 32-channel head coil. Functional data used 2D gradient-echo EPI (TR = 1.18 s). Structural data were acquired using an MP2RAGE sequence. Prior to preprocessing, MP2RAGE images underwent background noise suppression. Preprocessing was performed using fMRIPrep 21.0.1 [46], including motion correction and coregistration to the T1w image; slice-timing

correction was skipped. Post-processing steps were implemented in custom Python scripts, comprising nuisance regression of 22 confounds (6 motion parameters with derivatives, framewise displacement, global signal with derivative, CSF, white matter, and 5 anatomical CompCor components). High-pass filtering (> 0.0078 Hz) was implemented via discrete cosine transform basis functions during regression, followed by low-pass filtering (< 0.1 Hz) using a Savitzky-Golay filter. Full acquisition and preprocessing parameters are provided Supplementary Methods. Mean frame-wise displacement per story/run is shown in Figure S1.

Regions of interest

Regions of interest were anatomically defined from the probabilistic Harvard–Oxford cortical and subcortical atlas [33], split into left/right components and transformed to native functional space for signal extraction.

ROI groups used in summary figures

To summarize replicability across anatomically related regions, we organized Harvard–Oxford ROIs into two *a priori* sets: the peri-Sylvian language network (LANG) and the default-mode/parietal-integration network (DMN–PI), as specified below. These sets were used only for aggregation and display.

Language network (peri-Sylvian; LANG)

This set comprises bilateral peri-Sylvian speech/language cortex, specifically the following Harvard–Oxford regions: Heschl's gyrus (including H1/H2), planum temporale, planum polare, superior temporal gyrus (anterior and posterior divisions), middle temporal gyrus (anterior, posterior, and temporo-occipital divisions), inferior frontal gyrus (pars opercularis and pars triangularis), the frontal/central/parietal operculum, supramarginal gyrus (anterior and posterior), insular cortex, and temporal pole.

Default-mode/Parietal-integration network (DMN–PI)

This set comprises core default-mode hubs together with dorsal/posterior parietal regions that support long-timescale integration and integrative attention, namely: angular gyrus, precuneus cortex, posterior cingulate (Harvard–Oxford "Cingulate Gyrus, posterior division"), medial prefrontal/paracingulate cortex (Harvard–Oxford "Frontal Medial Cortex" and "Paracingulate Gyrus"), and superior parietal lobule. We use the label *DMN–PI*-rather than "DMN"-to make explicit that superior parietal lobule (a dorsal attention/control region) is included by design.

Voxelwise encoding models (overview)

For each voxel, we fit a ridge-regularized FIR encoding model mapping lagged features to BOLD responses on concatenated training stories. The held-out story remained untouched until evaluation. Hemodynamic delays were modeled with 14 lags (1.18–16.5 s). Model selection used bootstrap cross-validation on temporally contiguous chunks. Full design choices and hyperparameters are detailed in Supplementary Methods.

Evaluation and statistical inference (overview)

Predictive evidence was quantified as the correlation between held-out predictions and BOLD responses, with significance assessed at the region-of-interest (ROI) level using a block-permutation null and Simes omnibus test. For cross-story stability, we report the number of stories in which the ROI-level Simes test reached significance ($p < .05$), displayed on the cortical surface and within key anatomical ROIs. To map effects at the voxel level, we used cluster-based correction. Specifically, we regressed held-out BOLD responses onto the two predictions to isolate the unique contributions of shift and drift. The resulting statistical maps of mean coefficients, along with the directional contrast map ($\Delta\beta = \beta_{\text{shift}} - \beta_{\text{drift}}$), were thresholded using a cluster-defining threshold of $Z > 2.3$ and a cluster-extent significance level of $p < .05$. Voxelwise prevalence within ROIs was additionally assessed by calculating the proportion of voxels per ROI surviving the $Z > 2.3$ threshold. For detailed procedures, see Supplementary Methods.

Text-derived punctuation control (overview)

Mechanistically, event boundaries in speech often coincide with prosodic "edge" cues (pauses, pitch resets, durational changes) that elicit boundary-locked responses in bilateral auditory cortices, including Heschl's gyrus and STG [26, 37–39]. Because such prosodic edges often co-occur with punctuation, we also included a control analysis that tests whether an LLM-based, shift-like signal is reducible to punctuation/prosody. We constructed a punctuation regressor from the marks `.,:;` placed at the offsets of the corresponding words and evaluated a pairwise stage-2 model (unique *shift* vs. unique punctuation) on held-out stories. The construction and estimation details are provided in Supplementary Methods.

Timescale sweep for drift (overview)

To understand how long past information influences current processing, we systematically varied the integration timescale using a 'leak' parameter (ρ). This parameter controls the rate of decay: a low ρ means that past context fades quickly, while a high ρ allows information to persist over longer periods. We tested a wide range of values ($\rho \in 0.01, \ldots, 0.90$), re-calculating the drift signal and repeating the evaluation for each setting. We then summarized the results across regions by tracking how consistently the model predicted brain activity across different stories.

Declaration of competing interest

PH has received grants and honoraria from Novartis, Lundbeck, Mepha, Janssen, Boehringer Ingelheim, Neurolite, and OM Pharma outside of this work. No other disclosures were reported

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Ethics approval and consent to participate

The study was approved by the local ethics committee (KEK-ZH 2024-01314). All procedures complied with the Declaration of Helsinki. Written informed consent was obtained from the participant.

Data and code availability

All analysis code and derived regressors will be made available at https://github.com/homanlab/brainencode/.

Use of AI tools

An AI language model (ChatGPT, OpenAI) was used for language editing and clarity. It was not used to generate scientific content, perform analyses, or create figures or references. All text was reviewed and approved by the authors, who take full responsibility for the manuscript.

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Supplementary Information

Supplementary Methods

MRI acquisition and preprocessing (full)

Scanner: 7T Siemens MAGNETOM Terra; 32-channel head coil. Functional scans: 2D gradient-echo EPI, TR = 1.18 s; TE = 25 ms; flip angle 45°; voxel size $2.3 \times 2.3 \times 2.3\,\mathrm{mm}^3$ with 0.6 mm slice gap; multiband factor 3. Anatomy: T1-weighted multi-echo MPRAGE (TR = 6 s; TE = 1.99 ms; voxel size 0.68 mm isotropic). Fieldmaps: 2D gradient-echo (TR = 2.25 s; TE = 3.06 ms and 5.52 ms; flip angle 25°; matrix 64×64; slice thickness 4 mm).
For preprocessing, fMRIPrep [46] was used; standardized motion/distortion/slice-timing corrections and cross-run coregistration using FLIRT [47] to the first run of the session as reference were employed. Nuisance regression removed variance associated with six motion parameters, their first derivatives, and framewise displacement. Time series were low-pass filtered (fifth-order, zero-phase Butterworth, 0.1 Hz). Run-specific brain masks were applied, and data were voxelwise z-scored within run.

Text-audio alignment and resampling (full)

For forced alignment, we romanized/normalized the German text (lowercase, remove diacritics/punctuation) and applied the `torchaudio` MMS aligner [43, 48] to obtain Praat word start/end times. For each story, a *DataSequence* stored the word sequence, word timestamps, and the run-specific TR grid; timestamps were shifted by the measured audio onset. Resampling to TR was obtained by forming continuous-time feature traces by convolving impulse trains at word times with a Lanczos (sinc-windowed) interpolation kernel [49].

$$L_a(x) = \begin{cases} \mathrm{sinc}(x)\ \mathrm{sinc}\big(x/a\big), & |x| < a, \\ 0, & \text{otherwise}, \end{cases}$$

with window parameter a=1 for shift and a=3 for drift, and sampled at TR centers.

LLM features and signals (full)

Model and readout: Decoder-only LLM: *Llama-3.3-70B-Instruct* (HuggingFace), greedy decoding, 8-bit loading via `BitsAndBytes` [50]. Hidden states were read from layer L=79 using sliding token windows. For boundary log-probabilities, the look-back window was K=512 tokens. For base features (hidden states) we used a one-word stride with a variable look-back that ramps from 256 to 512 tokens and resets, amortizing compute (an engineering choice; longer discourse dependencies are governed by the drift integration below).
Drift: Define hidden state $\mathbf{H}_t$, leaky integration $\mathbf{g}_t = \rho\,\mathbf{g}_{t-1} + (1-\rho)\,\mathbf{H}_t$ with $\mathbf{g}_0 = \mathbf{H}_0$ and $\rho \in (0, 1)$; drift magnitude $\delta_t = \|\mathbf{g}_t - \mathbf{g}_{t-1}\|_2$. The per-TR drift series was z-scored within story after trimming and before FIR expansion. Unless otherwise stated, ρ=0.30.
Shift (boundary likelihood): We prompted the LLM to copy the text and insert a single special marker "¶" only at event boundaries [32]. At each word i we computed $\ell_i = \log p(\P \mid \mathrm{context}_{\leq i})$. Because boundary scoring used punctuated text while

alignment used normalized text, we reconciled positions with a sequence matcher and linearly interpolated unmatched gaps.
Boundary feature bank: From the per-TR series $\ell(t)$ we built four channels: (i) raw log-probability $x_1(t) = \ell(t)$; (ii) half-wave derivative $x_2(t) = \max\{\ell(t) - \ell(t-1), 0\}$; (iii) moving-average high-pass $x_3(t) = \ell(t) - (\ell * k_W)(t)$ with rectangular kernel k_W of width W=12 TRs; (iv) smoothed thresholded $x_4(t) = (\mathcal{G}_\sigma * r)(t)$ with σ=1 TR and $r(t) = \max\{\ell(t) - \tau, 0\}$, τ the 90th percentile of $\ell(t)$. Each channel was z-scored within story after trimming.
FIR expansion: For both drift (scalar) and shift (4-vector), we stacked D=14 lags (1.18–16.5 s), omitting lag 0.

Encoding model (full)

Design and lags: FIR with D=14 lags (covering the canonical positive BOLD lobe and early return to baseline). Lag 0 was omitted to enforce causality and guard against residual alignment error.

Estimation: Per-voxel ridge:

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \alpha\|\mathbf{w}\|_2^2,$$

with $\alpha \in \{0\} \cup \{10^{-6}, \ldots, 10^{0}\}$ (49 log-spaced values). Model selection used bootstrap cross-validation preserving temporal contiguity (chunk length 40 TRs; typical validation fraction ≈25%). Alphas were selected by maximizing validation correlation. The training design concatenated all training stories after trimming; validation blocks could span story boundaries.

Evaluation and inference (full)

Held-out predictive evidence (marginal)

For each voxel v and test story s, we computed Pearson's correlation $r_{v,s}$ between the held-out prediction and the measured BOLD time series. To obtain a null that preserves temporal autocorrelation, we randomly permuted the order of contiguous 10-TR chunks of the test-story BOLD response (block permutation; B=500 shuffles), recomputing the correlation each time. The empirical two-sided voxelwise p-value was

$$p_{v,s} \;=\; \frac{1 + \sum_{b=1}^{B} \mathbf{1}\big(\,|r_{v,s}^{(b)}| \geq |r_{v,s}|\,\big)}{B+1}.$$

Within each ROI R, voxelwise p-values $\{p_{v,s} : v \in R\}$ were combined using Simes' omnibus test to yield an ROI-level $p_{\text{Simes}}(R, s)$.

Cross-story stability (ROI Simes counts)

To quantify replicability across stories, we computed, for each ROI R, the *Simes count*

$$C(R) \;=\; \sum_{s=1}^{13} \mathbf{1}\big(p_{\text{Simes}}(R, s) < 0.05\big).$$

These counts summarize in how many held-out stories an ROI showed significant predictive evidence. They are descriptive of *replicable goodness-of-fit* (presence of an effect) and do not encode effect sign or voxelwise spatial extent. No additional voxelwise FDR is applied to the counts. Under the global null, $\mathbb{E}[C(R)] = 13 \times 0.05 = 0.65$.

Unique-effects estimation (stage-2) and directional contrast

To isolate independent contributions of the two predictors, we regressed the measured BOLD onto the two held-out predictions for each voxel v and story s:

$$y_{v,s}(t) = \beta_{v,s,\text{shift}}\, \hat{y}^{\text{shift}}_{v,s}(t) + \beta_{v,s,\text{drift}}\, \hat{y}^{\text{drift}}_{v,s}(t) + \varepsilon_{v,s}(t) \tag{S1}$$

All time series were z-scored *within story*; the intercept was omitted (expected ≈ 0). Coefficients were estimated by ordinary least squares. For each voxel, we then tested across stories whether the mean coefficient differed from zero (two-sided one-sample t-test, df= 12) separately for β_{shift} and β_{drift}. Voxelwise p-values were converted to z for display and controlled with BH–FDR at q=0.05 across cortical voxels. ROI-level inference for unique effects used Simes on the voxelwise p-values within each ROI.
We also formed the directional contrast $\Delta\beta_{v,s} = \beta_{v,s,\text{shift}} - \beta_{v,s,\text{drift}}$ and analyzed it analogously (across-story one-sample t, voxelwise BH–FDR q=0.05, plus ROI-level Simes).

Constants and implementation notes

Block-permutation chunk length for the test story was 10 TRs; the number of permutations was B=500. All tests were two-sided. BH–FDR control at q=0.05 was applied independently for each voxelwise map (β_{shift}, β_{drift}, and $\Delta\beta$). Simes tests were applied within ROIs to the relevant voxelwise p-values. No additional cluster-extent threshold was imposed on the voxelwise maps; small but FDR-significant clusters are therefore expected. The Simes tests provide ROI-level inference by combining voxelwise p-values within each anatomical ROI and do not implement a separate cluster-size correction.

Controls and parameter sweep (full)

Punctuation control: From the punctuated text we flagged tokens ending in `.,:;`. Flags were time-stamped at the *offset* of the preceding aligned word, converted to a per-TR series, z-scored within story, and expanded into a 14-lag FIR. Stage-1 ridge produced held-out predictions. Stage-2 unique-effects estimation used the pairwise model

$$y(t) = \beta_{\text{shift}}\hat{y}_{\text{shift}}(t) + \beta_{\text{punct}}\hat{y}_{\text{punct}}(t) + \varepsilon(t),$$

tested across stories as in above.
Timescale sweep: For $\rho \in \{0.01, 0.05, 0.10, \ldots, 0.90\}$ we recomputed δ_t, rebuilt the design, and repeated evaluation. For each ROI we recorded the number of test stories with Simes p<.05. We summarize (*i*) raw sums across ROIs and (*ii*) size-weighted mean significant stories per ROI (voxel-weighted) within LANG and DMN–PI.

Validating the LLM event-boundary metric

Rationale

Our boundary regressor is the LLM's *log-probability* of inserting a special boundary token at each word position. If this quantity truly captures event structure, then words labeled as event boundaries by humans should, on average, receive *higher* scores than non-boundaries, and a threshold on the score should recover human boundaries with few false alarms. We therefore evaluated the regressor with complementary, threshold-free and discrete criteria.

Data and alignment

All validation analyses in this appendix were conducted on a *separate* German crime story that we generated with GPT-4. We prompted the model to create a narrative with *clear, well-delimited events* to obtain unambiguous candidate boundaries. Nine fluent German raters (R1–R9) then read the text and independently inserted an event marker immediately *after* the last word of any event they perceived as ending. Raters were blinded to the LLM boundary scores and to each other's markings. This stimulus was *not* used for any fMRI modeling or evaluation and serves purely to test whether the LLM-derived boundary metric behaves as intended on a controlled narrative. We aligned scores to a word-time axis using a nominal reading rate (150 wpm), no audio or fMRI data were used for this appendix analysis, and computed the boundary score $\ell_i = \log p(\P \mid \text{context}_{\leq i})$ at every word. Unless stated otherwise, analyses in this appendix use tolerance = ±0 words around human boundaries, block bootstrap with 50 word blocks for CIs, and $n_{\text{boot}} = 10{,}000$ resamples.

Evaluation strategy

(1) Threshold-free discrimination

We treated the per-word score as a continuous decision variable and computed AUROC and AUPRC. AUROC reports the probability that a random human boundary receives a higher score than a random non-boundary; AUPRC reports precision vs. recall under class imbalance. We also report d' derived from AUROC via the equal-variance SDT link $d' = \sqrt{2}\,\Phi^{-1}(\text{AUROC})$ and, secondarily, a parametric d' from the positive/negative score distributions.

(2) Consensus sensitivity

To relate model scores to annotation strength, we recomputed AUROC while requiring that positives be marked by at least k raters (≥1, ≥2, ...). We show 95% CIs via block bootstrap and the count of positives at each threshold.

Results for the synthetic validation story

For our synthetic story (1,256 words; positives 49; base rate 0.039), we obtained:

- **AUROC = 0.9448** (95% CI **0.8991–0.9751**; 50 word blocks, $n_{\text{boot}} = 10{,}000$), corresponding to $\boldsymbol{d'}$ (from AUC) = **2.257**.
- **AUPRC = 0.4283** versus a class-balance baseline of **0.0390** (∼11× lift).

- **Distributional** d' (equal-variance SDT) = **2.849**.
- **Consensus link**: Spearman ρ between score and number of raters = **0.299**; AUROC remains $\gtrsim 0.95$ for mid-to-high consensus thresholds (counts per threshold are shown below).

These converging results demonstrate that the regressor is highly informative about human boundaries without manual thresholds. Together, they verify that the boundary metric behaves exactly like it is supposed to.

Control for punctuation

A potential confound is that event boundaries often coincide with sentence punctuation; if so, our boundary regressor might merely reflect punctuation. We therefore quantified (i) the association between punctuation and the LLM score, (ii) how well punctuation alone predicts human event boundaries, and (iii) whether the LLM score remains predictive *beyond* punctuation,both *within punctuation strata* and after removing the punctuation *mean-level shift*.

Operationalization

We flagged a word as "punctuation" if, after stripping closers (quotes/brackets), it ended in one of {. ! ? ... ; : ,}. Using the same word grid as the validation, we compared this indicator to the LLM boundary score $\ell_i = \log p(\P \mid \text{context}_{\leq i})$ and to the human event-boundary labels (union across raters; tolerance = ±0 words).

Mean-level shift

We observed that punctuation words have a higher *average* LLM score than non-punctuation words (here, means −7.29 vs. −12.93). This global offset is the *mean-level shift*. To test whether discrimination is driven only by this offset, we *residualized* the score by centering within groups: $s_{\text{resid}} = s - \mu_g$ with $g \in \{\text{non-punct}, \text{punct}\}$ and μ_g the mean score in group g. Residualizing removes the global two-mean difference; any remaining separation reflects *within-group* information.

Within punctuation strata

We also assessed discrimination *within* each punctuation subset, i.e., among *only* non-punctuation words and among *only* punctuation words. This conditions on punctuation status so that punctuation itself cannot explain any separation. If the LLM score still distinguishes human EBs in either subset, it carries information *beyond* punctuation.

Results

Punctuation words comprised 17.4% of tokens (218/1256) and had higher scores (means −7.29 vs. −12.93), with a point-biserial correlation $r = 0.604$ and AUROC (LLM score separating punctuation vs. non-punctuation) = 0.858, confirming a sensible association. As a baseline predictor of human boundaries, *punctuation alone* achieved AUROC = 0.781, AUPRC = 0.117 against a prevalence baseline = 0.039, and $d'_{\text{AUC}} = 1.10$; informative but markedly weaker than the LLM score (AUROC = 0.945, AUPRC = 0.428, $d'_{\text{AUC}} = 2.26$). Crucially, the LLM score remained predictive *within* punctuation

strata (AUROC = 0.929 among non-punctuation words; AUROC = 0.858 among punctuation-only words) and after removing the mean-level shift (residual AUROC = 0.917). Thus, while punctuation correlates with boundary likelihood, as expected, the LLM-derived boundary metric captures additional event structure that is not reducible to punctuation.

Supplementary Tables

Table S1: Cross-story region of interest significance counts. Number of stories (out of 13) in which region reached Simes $p<0.05$ for the *drift* predictor ($\rho = 0.3$) and for the *boundary* predictor. Hem = Hemisphere

ROI	Hem	Drift ρ=0.3	Boundary
Planum Temporale	Left	1	13
Superior Temporal Gyrus (anterior)	Left	1	13
Heschl's Gyrus (H1/H2)	Left	0	13
Planum Polare	Left	0	13
Superior Temporal Gyrus (anterior)	Right	0	13
Superior Temporal Gyrus (posterior)	Left	0	13
Superior Temporal Gyrus (posterior)	Right	4	12
Parietal Operculum Cortex	Left	2	12
Heschl's Gyrus (H1/H2)	Right	1	12
Middle Temporal Gyrus (anterior)	Left	1	11
Planum Polare	Right	0	11
Central Opercular Cortex	Left	1	10
Middle Temporal Gyrus (anterior)	Right	0	9
Supramarginal Gyrus (posterior)	Left	4	9
Supramarginal Gyrus (posterior)	Right	2	9
Planum Temporale	Right	1	9
Middle Temporal Gyrus (posterior)	Right	2	8
Middle Temporal Gyrus (temporo-occipital)	Left	3	8
Middle Temporal Gyrus (posterior)	Left	0	7
Inferior Frontal Gyrus (pars triangularis)	Right	2	7
Supramarginal Gyrus (anterior)	Left	4	4
Middle Temporal Gyrus (temporo-occipital)	Right	1	3
Lateral Occipital Cortex (superior)	Left	2	3
Intracalcarine Cortex	Right	1	3
Supplementary Motor Area (Juxtapositional Lobule)	Left	3	3
Precuneus Cortex	Left	4	3
Precuneus Cortex	Right	4	3
Supracalcarine Cortex	Left	2	3
Middle Frontal Gyrus	Right	2	3
Precentral Gyrus	Left	1	3
Superior Parietal Lobule	Left	4	2
Superior Parietal Lobule	Right	2	2
Angular Gyrus	Right	2	2
Intracalcarine Cortex	Left	2	2
Cuneal Cortex	Left	2	2
Cuneal Cortex	Right	1	2
Occipital Fusiform Gyrus	Right	1	2
Central Opercular Cortex	Right	1	2
Parietal Operculum Cortex	Right	3	2
Supracalcarine Cortex	Right	2	2
Occipital Pole	Left	0	2
Occipital Pole	Right	2	2
Middle Frontal Gyrus	Left	2	2
Temporal Pole	Left	0	2
Temporal Pole	Right	2	2
Postcentral Gyrus	Left	1	1
Supramarginal Gyrus (anterior)	Right	3	1
Angular Gyrus	Left	6	1
Lateral Occipital Cortex (superior)	Right	3	1
Lateral Occipital Cortex (inferior)	Left	2	1
Medial Frontal Cortex	Right	1	1
Supplementary Motor Area (Juxtapositional Lobule)	Right	2	1
Posterior Cingulate Gyrus	Left	1	1
Lingual Gyrus	Left	1	1
Lingual Gyrus	Right	1	1
Superior Frontal Gyrus	Left	2	1
Occipital Fusiform Gyrus	Left	4	1
Inferior Frontal Gyrus (pars triangularis)	Left	3	1

Continued on next page

ROI	Hem	Drift ρ=0.3	Boundary
Inferior Frontal Gyrus (pars opercularis)	Right	3	1
Precentral Gyrus	Right	1	1
Amygdala	Left	0	0
Accumbens	Left	0	0
Inferior Temporal Gyrus (anterior)	Left	0	0
Inferior Temporal Gyrus (anterior)	Right	0	0
Right Lateral Ventricle		0	0
Inferior Temporal Gyrus (posterior)	Left	0	0
Inferior Temporal Gyrus (posterior)	Right	0	0
Thalamus	Right	0	0
Inferior Temporal Gyrus (temporo-occipital)	Left	0	0
Inferior Temporal Gyrus (temporo-occipital)	Right	1	0
Caudate	Right	0	0
Postcentral Gyrus	Right	2	0
Putamen	Right	1	0
Hippocampus	Right	0	0
Frontal Pole	Left	1	0
Frontal Pole	Right	1	0
Amygdala	Right	0	0
Accumbens	Right	0	0
Lateral Occipital Cortex (inferior)	Right	1	0
Medial Frontal Cortex	Left	1	0
Subcallosal Cortex	Left	3	0
Subcallosal Cortex	Right	3	0
Paracingulate Gyrus	Left	1	0
Paracingulate Gyrus	Right	1	0
Anterior Cingulate Gyrus	Left	1	0
Anterior Cingulate Gyrus	Right	1	0
Insular Cortex	Left	0	0
Insular Cortex	Right	1	0
Posterior Cingulate Gyrus	Right	4	0
Cingulate Gyrus (posterior)	Right	1	0
Frontal Orbital Cortex	Left	0	0
Frontal Orbital Cortex	Right	0	0
Parahippocampal Gyrus (anterior)	Left	0	0
Parahippocampal Gyrus (anterior)	Right	0	0
Parahippocampal Gyrus (posterior)	Left	2	0
Parahippocampal Gyrus (posterior)	Right	3	0
Lingual Gyrus		0	0
Temporal Fusiform Cortex (anterior)	Left	1	0
Temporal Fusiform Cortex (anterior)	Right	2	0
Temporal Fusiform Cortex (posterior)	Left	1	0
Temporal Fusiform Cortex (posterior)	Right	1	0
Temporal Occipital Fusiform Cortex	Left	1	0
Temporal Occipital Fusiform Cortex	Right	1	0
Superior Frontal Gyrus	Right	1	0
Occipital Fusiform Gyrus	Right	1	0
Frontal Operculum Cortex	Left	2	0
Frontal Operculum Cortex	Right	1	0
Central Opercular Cortex	Left	1	0
Central Opercular Cortex	Right	1	0
Parietal Operculum Cortex	Left	2	0
Parietal Operculum Cortex	Right	3	0
Heschl's Gyrus (H1/H2)	Right	1	0
Supracalcarine Cortex	Left	2	0
Supracalcarine Cortex	Right	2	0
Occipital Pole	Right	2	0
Thalamus	Left	0	0
Middle Frontal Gyrus	Left	2	0
Middle Frontal Gyrus	Right	2	0
Inferior Frontal Gyrus (pars triangularis)	Left	2	0
Caudate	Left	0	0
Inferior Frontal Gyrus (pars opercularis)	Left	2	0
Inferior Frontal Gyrus (pars opercularis)	Right	3	0
Putamen	Left	1	0

Continued on next page

ROI	**Hem**	**Drift** ρ=0.3	**Boundary**
Pallidum	Left	1	0
Precentral Gyrus	Left	1	0
Precentral Gyrus	Right	1	0
Temporal Pole	Left	0	0
Temporal Pole	Right	0	0
Hippocampus	Left	0	0
Superior Temporal Gyrus (anterior)	Left	1	0
Superior Temporal Gyrus (anterior)	Right	0	0

Table S2: Region of interest-level directional contrast. Mean $\overline{\Delta\beta} = \overline{\beta_{\text{shift}} - \beta_{\text{drift}}}$, Simes $p(\Delta\beta)$ across voxels, direction label (shift≥drift / drift≥shift / n.s.), and ROI size (n voxels). Full ROI names are used; hemisphere is listed separately. Hem = Hemisphere

ROI	Hem	$\overline{\Delta\beta}$	$p_{\text{Simes}}(\Delta\beta)$	Direction	n
Amygdala	Left	-0.000558	0.628836	n.s.	106
Superior Temporal Gyrus (posterior division)	Left	0.052598	4.7546e-07	shift≥drift	392
Superior Temporal Gyrus (posterior division)	Right	0.046555	6.1714e-07	shift≥drift	478
Accumbens	Left	0.000127	0.169430	n.s.	10
Middle Temporal Gyrus (anterior division)	Left	0.035144	4.2532e-06	shift≥drift	165
Middle Temporal Gyrus (anterior division)	Right	0.017205	1.2101e-05	shift≥drift	114
Middle Temporal Gyrus (posterior division)	Left	0.016087	2.9831e-05	shift≥drift	365
Middle Temporal Gyrus (posterior division)	Right	0.021461	2.4058e-06	shift≥drift	351
Middle Temporal Gyrus (temporooccipital part)	Left	0.011083	0.0054266	shift≥drift	341
Middle Temporal Gyrus (temporooccipital part)	Right	0.001549	0.00021484	shift≥drift	424
Inferior Temporal Gyrus (anterior division)	Left	0.005764	0.0189603	shift≥drift	89
Inferior Temporal Gyrus (anterior division)	Right	-0.000182	0.0074471	drift≥shift	92
Lateral Ventricle	Right	0.000043	0.0917890	n.s.	272
Inferior Temporal Gyrus (posterior division)	Left	0.004657	0.0314186	shift≥drift	327
Inferior Temporal Gyrus (posterior division)	Right	0.002930	0.0037946	shift≥drift	325
Thalamus	Right	0.003165	0.385500	n.s.	80
Inferior Temporal Gyrus (temporooccipital part)	Left	0.007973	0.0187952	shift≥drift	363
Inferior Temporal Gyrus (temporooccipital part)	Right	-0.003653	0.0297292	drift≥shift	342
Caudate	Right	-0.002201	0.651655	n.s.	45
Postcentral Gyrus	Left	-0.000610	0.0017277	drift≥shift	1362
Postcentral Gyrus	Right	-0.005481	0.0111283	drift≥shift	1274
Putamen	Right	-0.002019	0.0601635	n.s.	5
Superior Parietal Lobule	Left	-0.007055	0.0217556	drift≥shift	726
Superior Parietal Lobule	Right	-0.004071	0.0677372	n.s.	452
Hippocampus	Right	-0.003875	0.0126435	drift≥shift	262
Supramarginal Gyrus (anterior division)	Left	0.010508	0.0191072	shift≥drift	401
Supramarginal Gyrus (anterior division)	Right	-0.001669	0.0137826	drift≥shift	360
Frontal Pole	Left	0.002122	0.0993097	n.s.	3052
Frontal Pole	Right	0.004387	0.0212449	shift≥drift	3841
Amygdala	Right	-0.001834	0.188738	n.s.	149
Supramarginal Gyrus (posterior division)	Left	0.022588	5.1282e-07	shift≥drift	433
Supramarginal Gyrus (posterior division)	Right	0.025459	9.0287e-06	shift≥drift	725
Angular Gyrus	Left	0.009036	0.082200	n.s.	396
Angular Gyrus	Right	0.002227	4.8223e-05	shift≥drift	577
Accumbens	Right	0.002391	0.0596555	n.s.	7
Lateral Occipital Cortex (superior division)	Left	0.003412	0.0170109	shift≥drift	1929
Lateral Occipital Cortex (superior division)	Right	-0.001109	0.00075845	drift≥shift	1841
Lateral Occipital Cortex (inferior division)	Left	-0.006172	0.0118370	drift≥shift	858
Lateral Occipital Cortex (inferior division)	Right	0.000268	0.0225301	shift≥drift	779
Intracalcarine Cortex	Left	0.010502	0.479807	n.s.	190
Intracalcarine Cortex	Right	0.010586	0.0305649	shift≥drift	188
Medial Frontal Cortex	Left	0.003688	0.0330878	shift≥drift	117
Medial Frontal Cortex	Right	0.015079	0.0180565	shift≥drift	182
Supplementary Motor Area (Juxtapositional Lobule)	Left	0.006281	0.00020104	shift≥drift	310
Supplementary Motor Area (Juxtapositional Lobule)	Right	0.002684	0.00012647	shift≥drift	238
Subcallosal Cortex	Left	0.000831	0.0154952	shift≥drift	290
Subcallosal Cortex	Right	0.001144	0.0053657	shift≥drift	314
Paracingulate Gyrus	Left	0.003980	0.120625	n.s.	627
Paracingulate Gyrus	Right	0.001372	0.056431	n.s.	745
Anterior Cingulate Gyrus	Left	-0.000990	0.0304911	drift≥shift	452
Anterior Cingulate Gyrus	Right	-0.001212	0.046760	drift≥shift	518
Insular Cortex	Left	0.004344	0.0210726	shift≥drift	579
Insular Cortex	Right	0.006991	0.0451745	shift≥drift	507
Posterior Cingulate Gyrus	Left	0.001967	0.0397666	shift≥drift	537
Posterior Cingulate Gyrus	Right	0.001918	0.0033638	shift≥drift	499
Precuneus Cortex	Left	0.003812	0.0106011	shift≥drift	1174
Precuneus Cortex	Right	0.002250	0.0222826	shift≥drift	989
Cuneal Cortex	Left	-0.001689	0.0298652	drift≥shift	182

ROI	Hem	$\overline{\Delta\beta}$	$p_{\text{Simes}}(\Delta\beta)$	Direction	n
Cuneal Cortex	Right	-0.001986	0.0309927	drift≥shift	147
Frontal Orbital Cortex	Left	0.001413	0.0861123	n.s.	826
Frontal Orbital Cortex	Right	0.004478	0.0344690	shift≥drift	717
Parahippocampal Gyrus (anterior division)	Left	0.002226	0.0767000	n.s.	578
Parahippocampal Gyrus (anterior division)	Right	-0.004042	0.0026907	drift≥shift	621
Parahippocampal Gyrus (posterior division)	Left	0.003034	0.0075103	shift≥drift	328
Parahippocampal Gyrus (posterior division)	Right	0.000701	0.0116060	shift≥drift	287
Lingual Gyrus	Left	-0.000492	0.179905	n.s.	856
Lingual Gyrus	Right	0.004754	0.0090806	shift≥drift	827
Temporal Fusiform Cortex (anterior division)	Left	-0.000352	0.0808884	n.s.	178
Temporal Fusiform Cortex (anterior division)	Right	-0.004111	0.0150039	drift≥shift	157
Temporal Fusiform Cortex (posterior division)	Left	0.000391	0.0134519	shift≥drift	455
Temporal Fusiform Cortex (posterior division)	Right	-0.003932	0.142924	n.s.	332
Temporal Occipital Fusiform Cortex	Left	0.002801	0.0280758	shift≥drift	430
Temporal Occipital Fusiform Cortex	Right	0.003651	0.0289416	shift≥drift	518
Superior Frontal Gyrus	Left	0.005437	0.0554195	n.s.	1476
Superior Frontal Gyrus	Right	0.004250	0.00035849	shift≥drift	1213
Occipital Fusiform Gyrus	Left	-0.008533	0.0237896	drift≥shift	597
Occipital Fusiform Gyrus	Right	0.009780	0.147971	n.s.	674
Frontal Operculum Cortex	Left	0.012396	0.0077966	shift≥drift	213
Frontal Operculum Cortex	Right	0.008940	0.234855	n.s.	154
Central Opercular Cortex	Left	0.027159	5.6780e-06	shift≥drift	401
Central Opercular Cortex	Right	0.021387	6.3127e-05	shift≥drift	373
Parietal Operculum Cortex	Left	0.051082	7.9430e-06	shift≥drift	190
Parietal Operculum Cortex	Right	0.009672	1.2898e-04	shift≥drift	156
Planum Polare	Left	0.042650	7.7281e-06	shift≥drift	147
Planum Polare	Right	0.040514	1.6406e-06	shift≥drift	145
Heschl's Gyrus	Left	0.095693	7.2479e-06	shift≥drift	81
Heschl's Gyrus	Right	0.068474	5.5663e-06	shift≥drift	70
Planum Temporale	Left	0.091250	4.5656e-07	shift≥drift	97
Planum Temporale	Right	0.041144	1.9731e-05	shift≥drift	82
Supracalcarine Cortex	Left	0.009460	0.113525	n.s.	50
Supracalcarine Cortex	Right	0.019009	0.160583	n.s.	23
Occipital Pole	Left	-0.003617	0.0511000	n.s.	835
Occipital Pole	Right	0.002611	0.109661	n.s.	748
Thalamus	Left	0.004953	0.100008	n.s.	123
Middle Frontal Gyrus	Left	0.004531	1.2366e-04	shift≥drift	1180
Middle Frontal Gyrus	Right	0.005288	5.6822e-05	shift≥drift	1334
Inferior Frontal Gyrus (pars triangularis)	Left	0.002106	0.0022778	shift≥drift	365
Inferior Frontal Gyrus (pars triangularis)	Right	0.016964	0.0126339	shift≥drift	314
Caudate	Left	0.004568	0.0398659	shift≥drift	49
Inferior Frontal Gyrus (pars opercularis)	Left	0.003196	0.0100426	shift≥drift	331
Inferior Frontal Gyrus (pars opercularis)	Right	0.006965	0.0020086	shift≥drift	297
Putamen	Left	-0.007069	0.110458	n.s.	5
Pallidum	Left	-0.003162	0.115245	n.s.	3
Precentral Gyrus	Left	0.002591	5.9445e-05	shift≥drift	1674
Precentral Gyrus	Right	-0.001276	0.0132759	drift≥shift	1481
Temporal Pole	Left	0.006104	2.4259e-05	shift≥drift	1249
Temporal Pole	Right	0.006147	0.00022718	shift≥drift	1264
Hippocampus	Left	0.002323	0.0053110	shift≥drift	258
Superior Temporal Gyrus (anterior division)	Left	0.105594	3.1934e-07	shift≥drift	128
Superior Temporal Gyrus (anterior division)	Right	0.078284	6.5573e-08	shift≥drift	132

Table S3: Story corpus and scan schedule. Narrated stories (German titles with canonical/original titles where applicable), authors, and scan schedule. Durations are computed from TR counts ($TR = 1.18\,s$).

Canonical title	Author	Session/Run	Date and time	TRs	Minutes
La Main	Guy de Maupassant	S01 / R1	2024-04-16 16:20	686	13.5
The Adventure of the Blue Carbuncle	Arthur Conan Doyle	S02 / R1	2024-04-18 15:18	2726	53.6
Ligeia	Edgar Allan Poe	S02 / R2	2024-04-18 16:17	1781	35.0
The Musgrave Ritual	Arthur Conan Doyle	S03 / R1	2024-04-26 15:05	2765	54.4
King Pest	Edgar Allan Poe	S03 / R2	2024-04-26 16:03	1575	31.0
The Five Orange Pips	Arthur Conan Doyle	S04 / R1	2024-05-03 13:57	2555	50.2
The Premature Burial	Edgar Allan Poe	S04 / R2	2024-05-03 14:50	1904	37.4
MS. Found in a Bottle	Edgar Allan Poe	S06 / R1	2024-05-27 14:57	1458	28.7
The Black Cat	Edgar Allan Poe	S06 / R2	2024-05-27 15:28	1445	28.4
The Sunningdale Mystery	Agatha Christie	S06 / R3	2024-05-27 15:58	1361	26.8
The Stretelli Case	Edgar Wallace	S07 / R1	2024-05-29 14:26	1291	25.4
The Masque of the Red Death	Edgar Allan Poe	S07 / R2	2024-05-29 14:54	833	16.4
Die Pflanzen des Dr. Cinderella	Gustav Meyrink	S07 / R3	2024-05-29 15:12	1177	23.1

Supplementary Figures

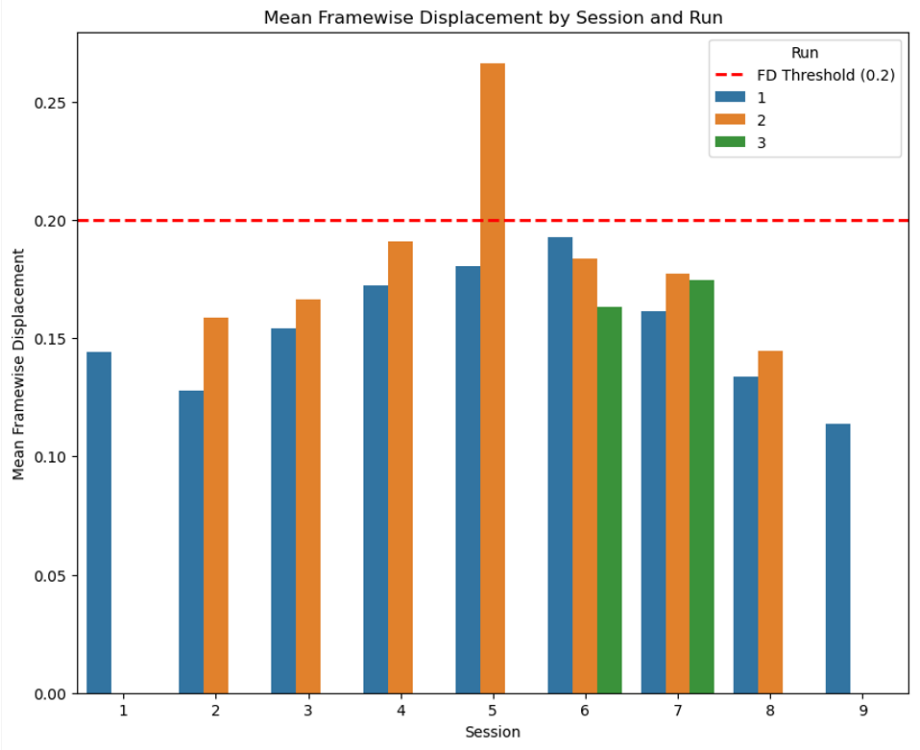


Figure S1: Mean frame-wise displacement in BOLD data.

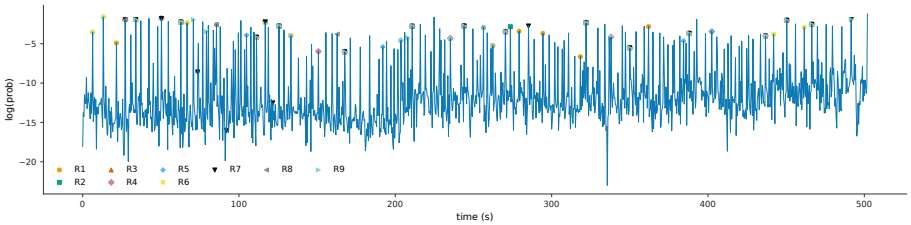
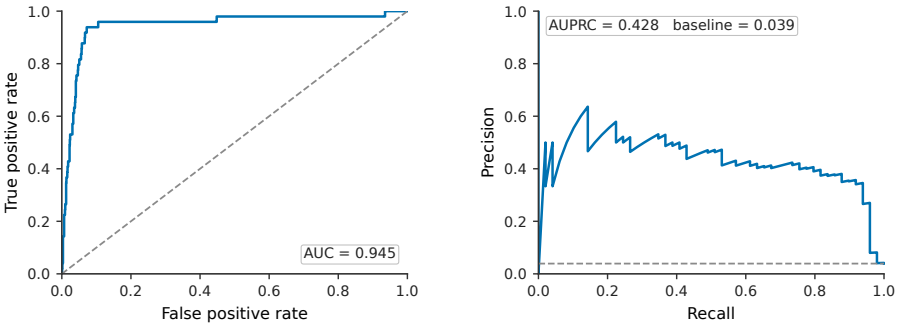
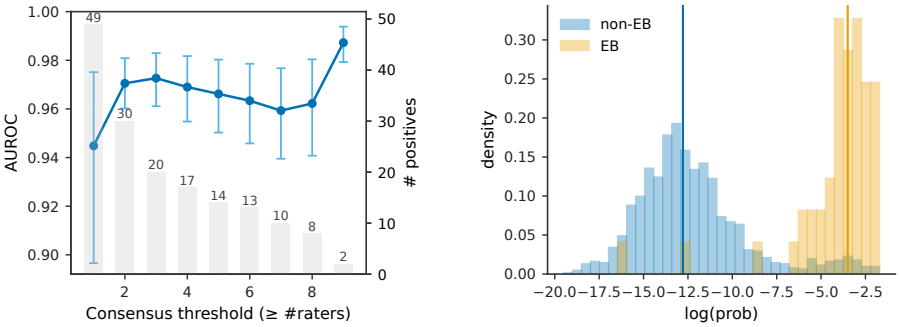


Figure S2: Boundary score over time (WORD level). The curve shows the per-word *log-probability* that the LLM would insert the boundary token ($\log p(\P \mid \text{context})$) aligned to word times. Colored markers indicate human event boundaries from individual raters (R1–R9), plotted at the word immediately preceding each annotated boundary.



(a) ROC (AUROC = 0.945; 95% CI 0.899–0.975). **(b) Precision–Recall (AUPRC = 0.428; baseline = 0.039).**

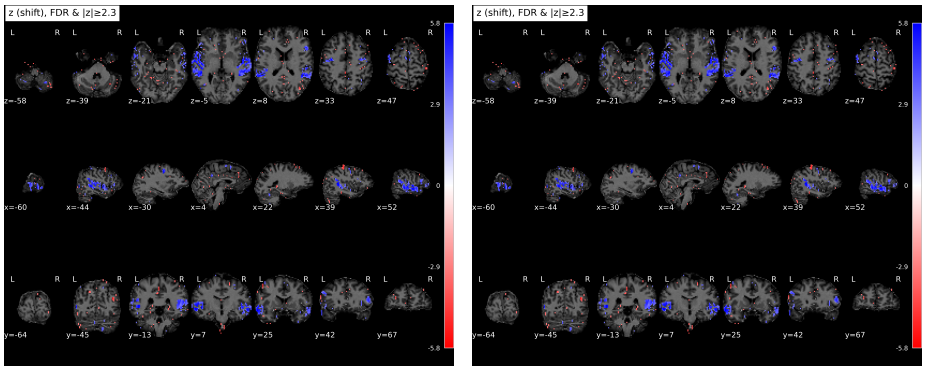
Figure S3: Threshold-free validation of the boundary metric. (**Left**) ROC illustrates separability of boundary vs. non-boundary words (area under the curve, AUROC). Confidence intervals are block-bootstrapped with 50 word blocks and $n_{\text{boot}} = 10{,}000$ to respect temporal autocorrelation. (**Right**) PR curve shows performance under class imbalance; the dashed line marks the class-balance baseline (positives = 49/1,256 words). Together, these plots quantify discrimination without choosing a detection threshold.



(a) Consensus sensitivity with 95% CIs and #positives.

(b) Score distributions for EB vs. non-EB words.

Figure S4: Calibration to human agreement and score separation. (**Left**) AUROC as the consensus threshold increases (≥#raters). Shaded bars (right axis) give the number of positives at each threshold, clarifying uncertainty when few positives remain. The metric stays highly discriminative across mid-to-high consensus levels. (**Right**) Histograms of the boundary score (log-prob) for positive vs. negative words with median markers; the parametric SDT effect size across the two distributions is $d' = 2.85$.



(a) Unique shift (vs. punctuation) **(b) Unique punctuation (vs. shift)**

Figure S5: Punctuation control: voxelwise unique-effects maps. Across-story one-sample $t \rightarrow z$ maps (df= 12) for the pairwise stage-2 model $y(t) = \beta_{\text{shift}}\hat{y}_{\text{shift}}(t) + \beta_{\text{punct}}\hat{y}_{\text{punct}}(t) + \varepsilon(t)$. Maps are BH–FDR masked (q=0.05) with a display threshold of $|z| \geq 2.3$ under a symmetric diverging colormap. **Unique shift** shows robust, bilateral clusters along the peri-Sylvian belt (Heschl's gyrus, planum temporale/polare, STG/MTG), whereas **unique punctuation** is markedly weaker and sparser in these same regions, indicating that boundary-linked responses are not reducible to pauses/typography.